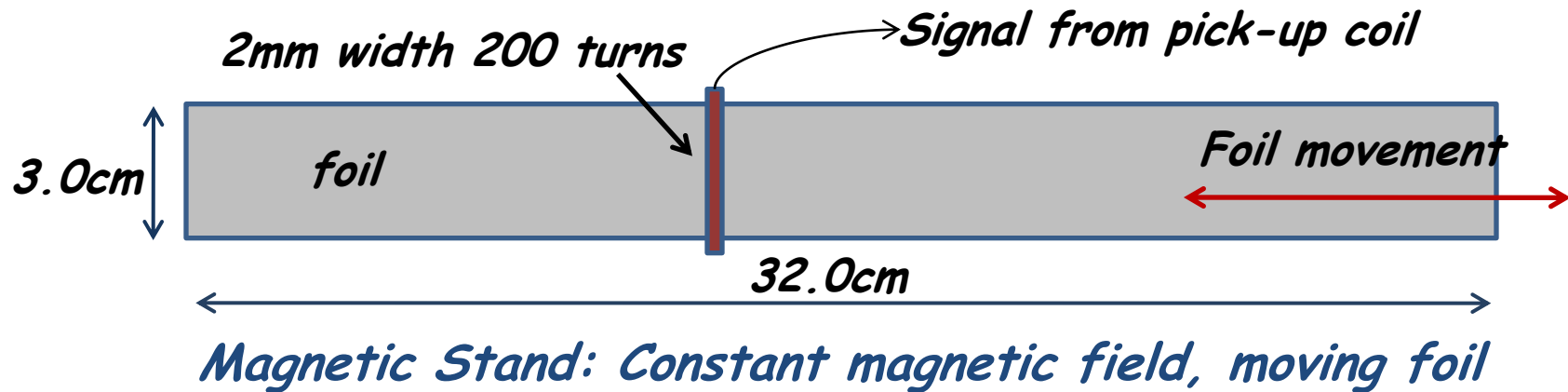
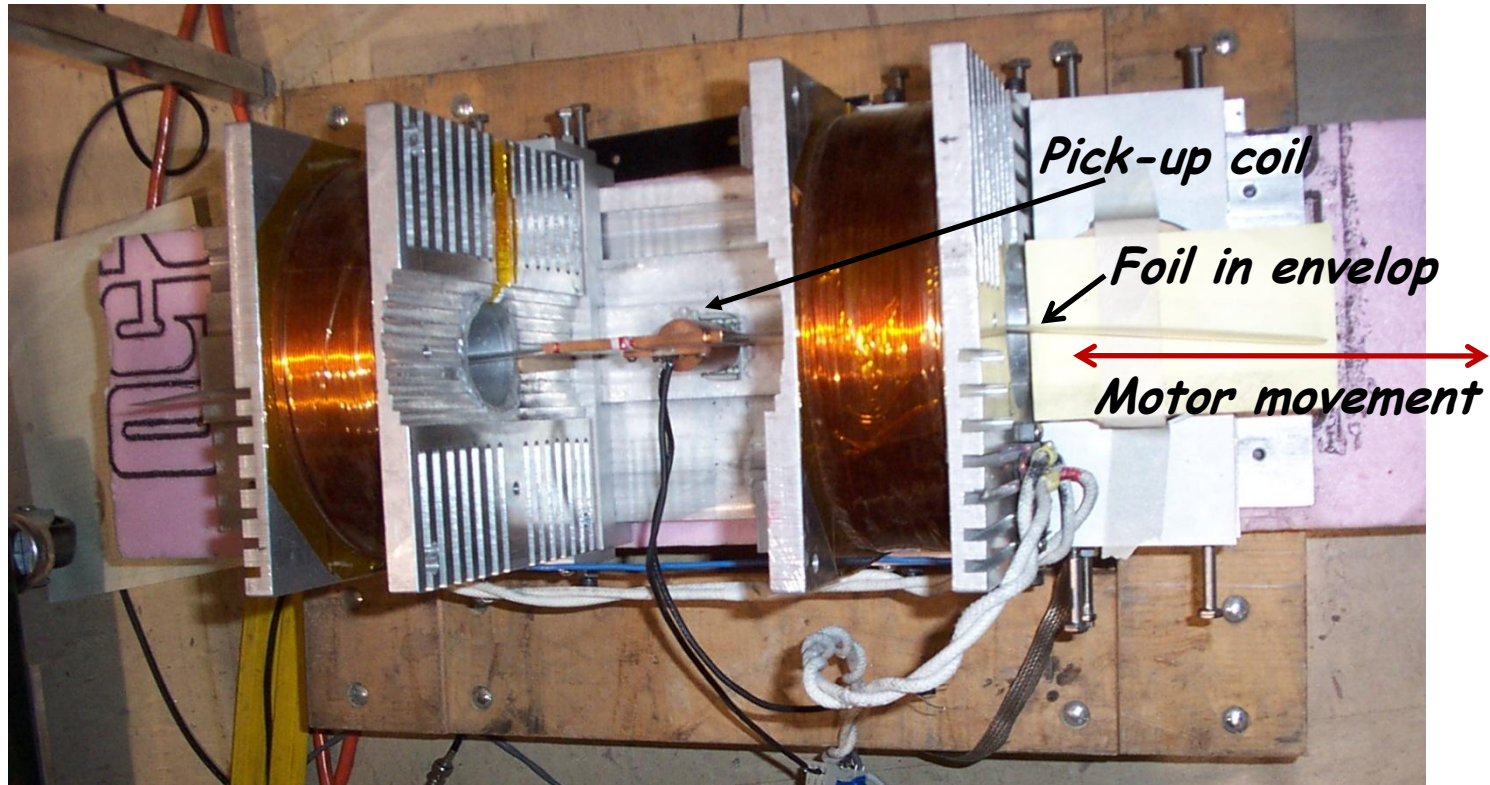


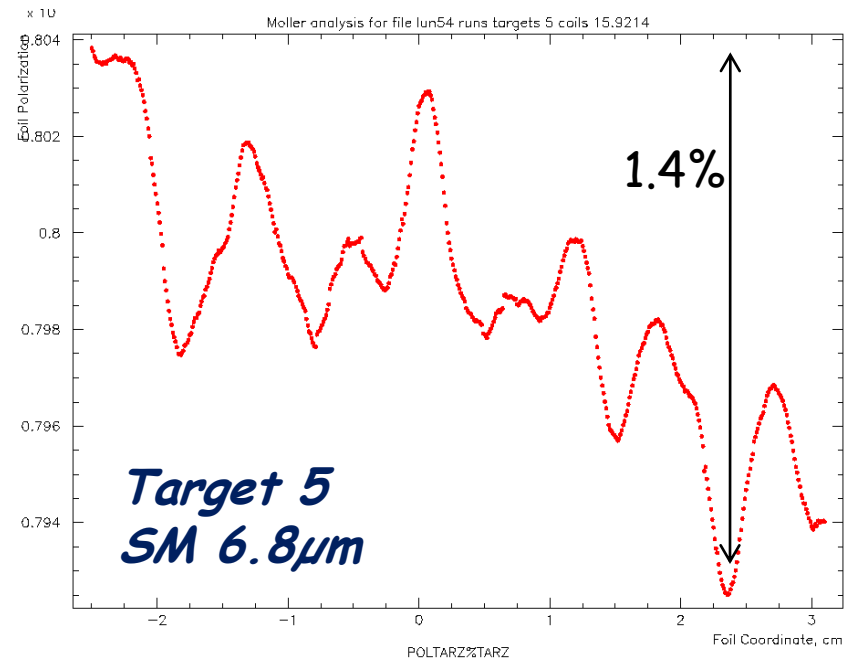
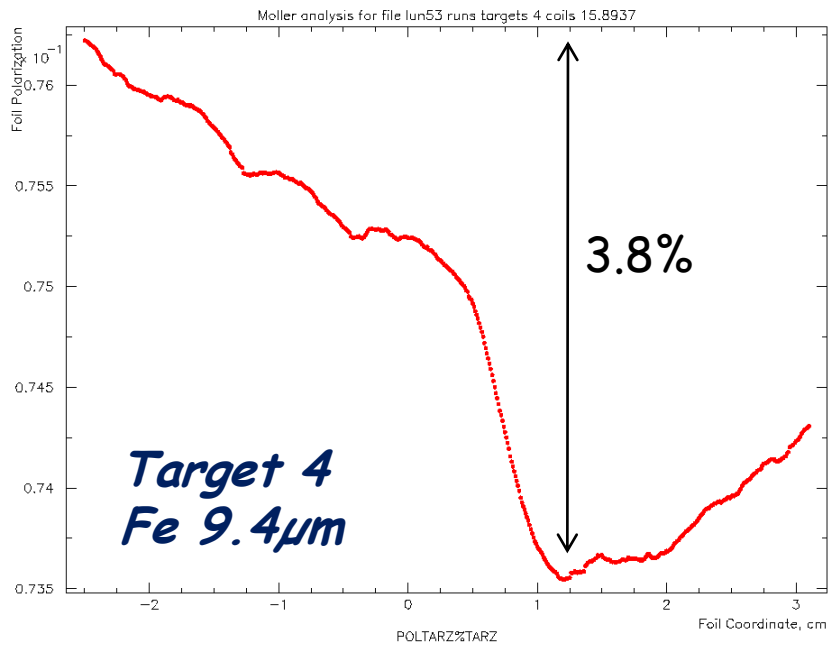
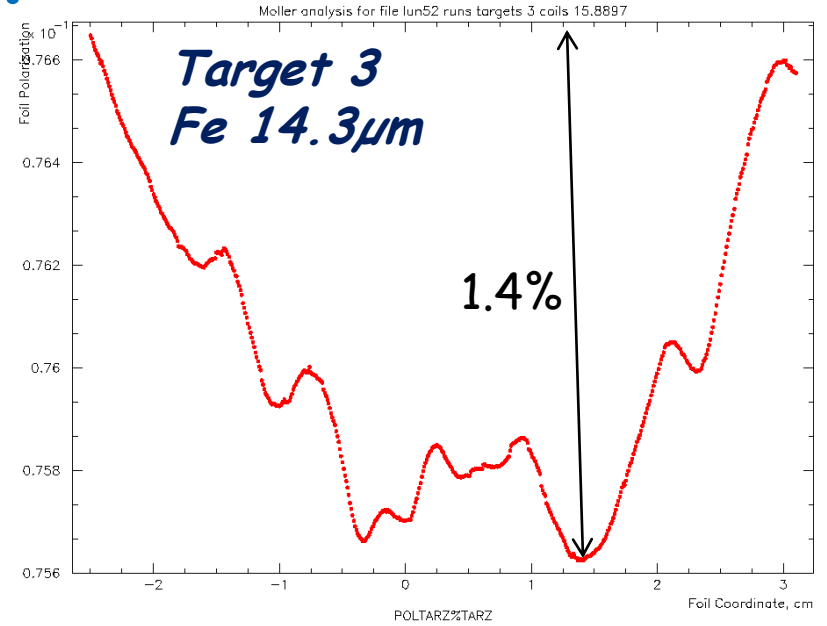
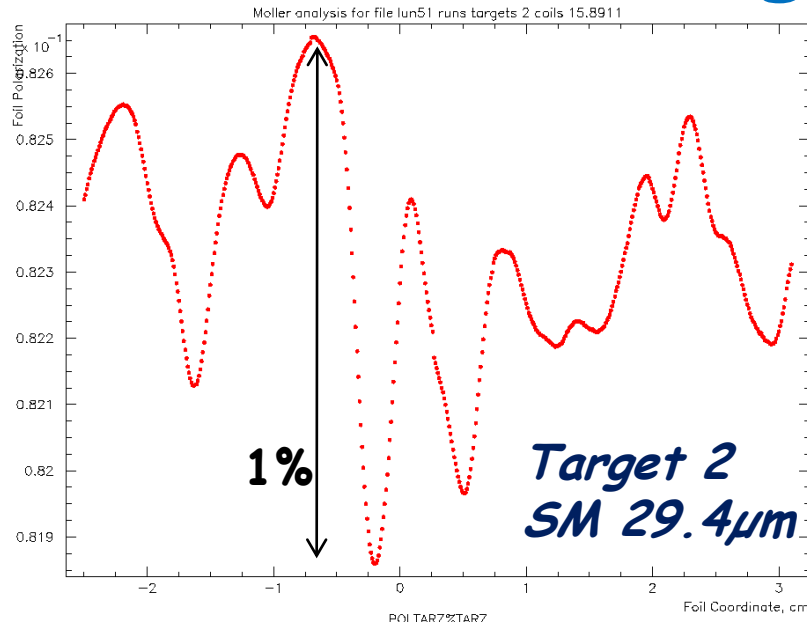
Old Møller Target

Systematic Error

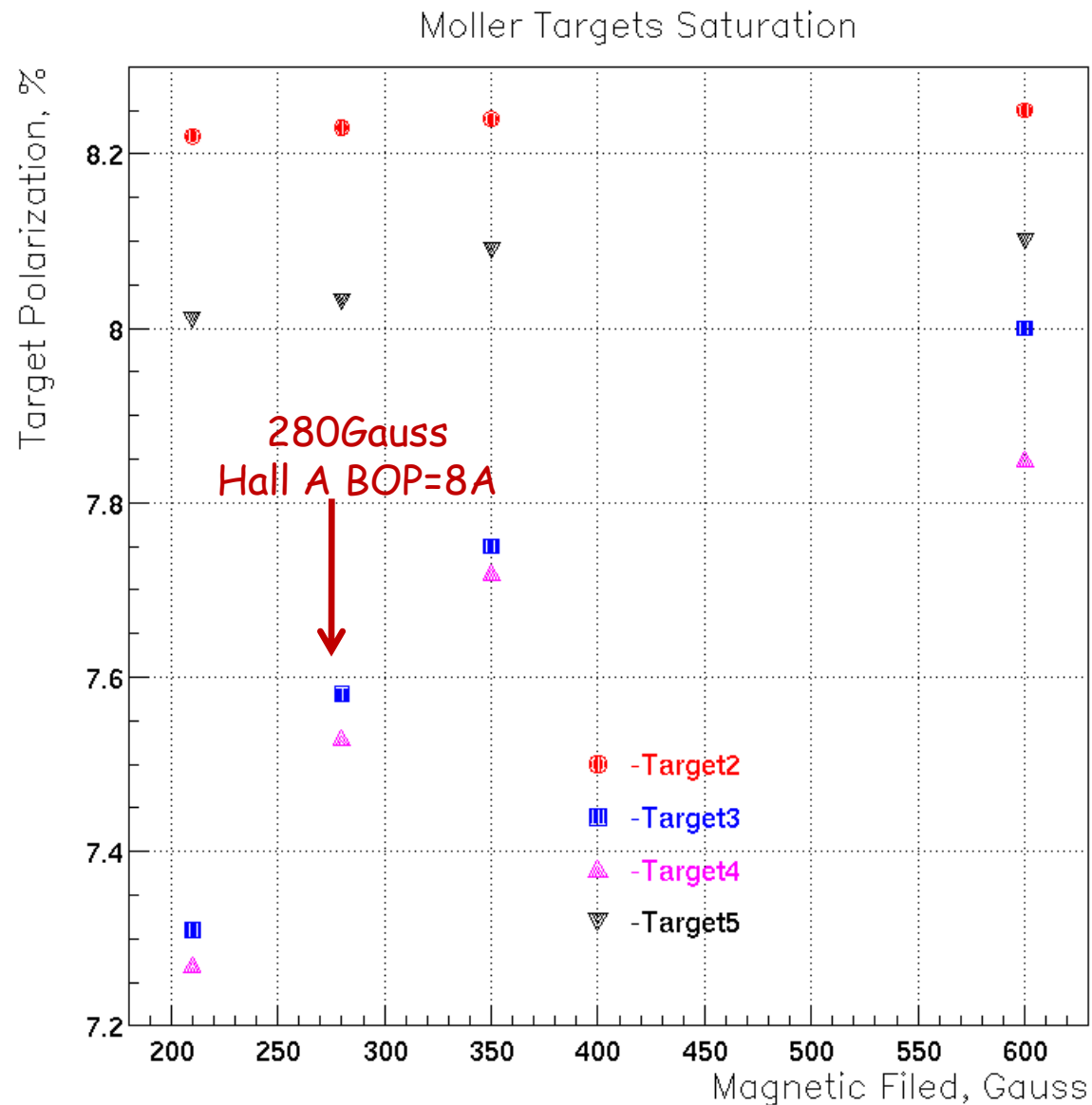
Foil polarization measurement



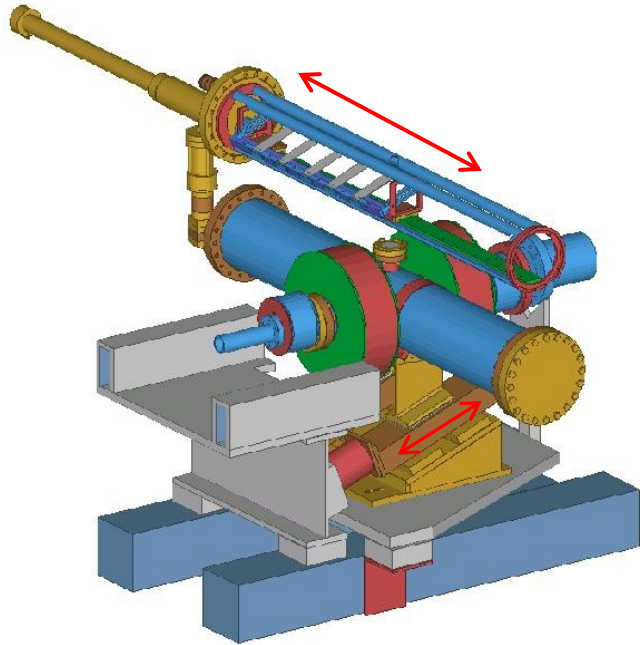
Møller targets polarization



Møller Targets Saturation



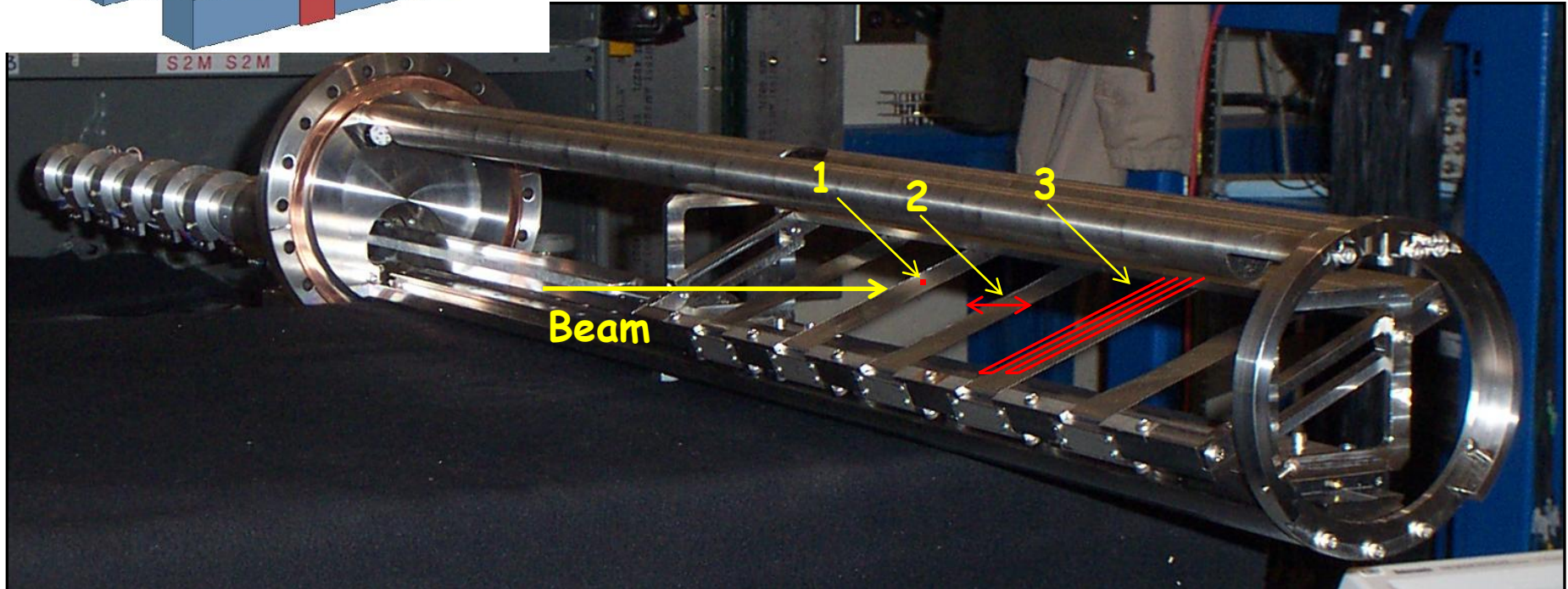
Types of beam polarization measurement



Type 1 - measurement in the point

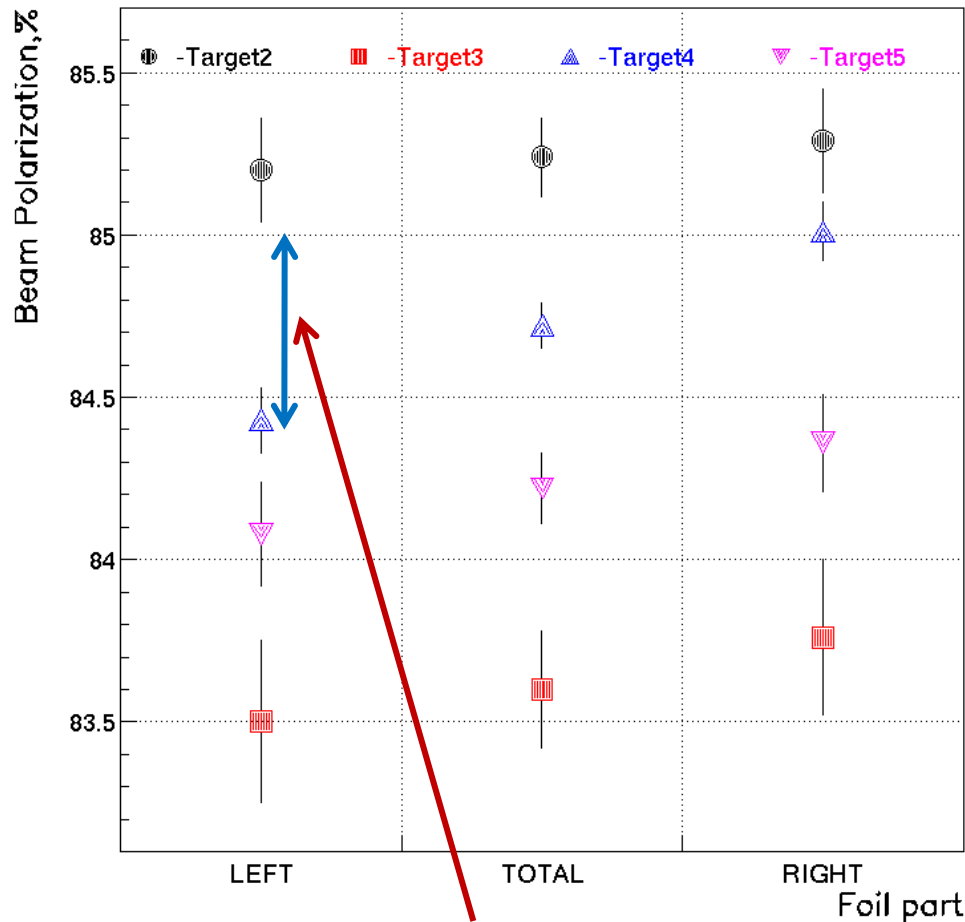
Type 2 - foil scan in transverse direction

Type 3 - double direction foil scan (like snake)



Double scan (type 3) results

Moller Targets Double Scan Results



$$85.01/84.46=0.65\%$$

$$7.675/7.542=1.76\%$$

LEFT - integral polarization
-2.4cm $\leftrightarrow$ 0.1cm

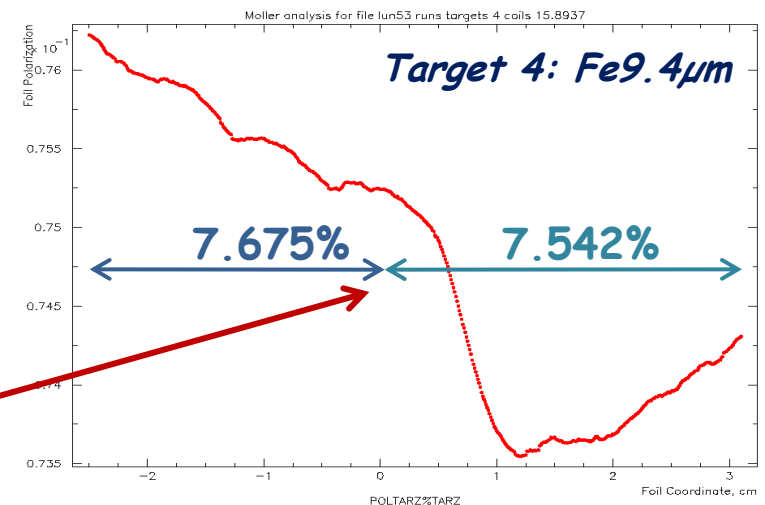
RIGHT - integral polarization
-0.1cm $\leftrightarrow$ 2.9cm

TOTAL - integral polarization
-2.4 cm $\leftrightarrow$ 2.9cm

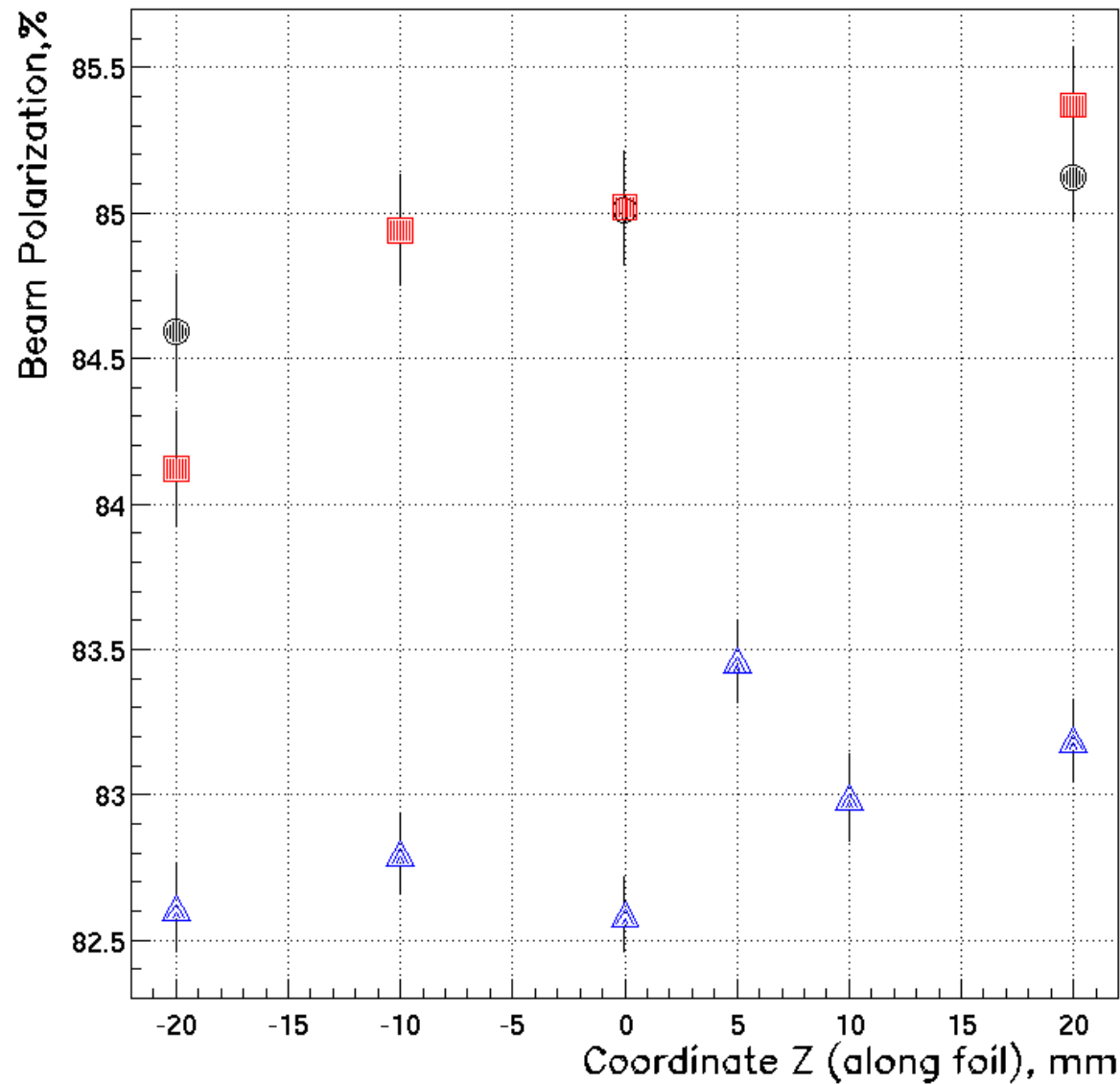
Available foil scan along the beam direction is:

-2.5 cm $\leftrightarrow$ 3.0cm

Syst. Error for target 4:
+/-0.3%

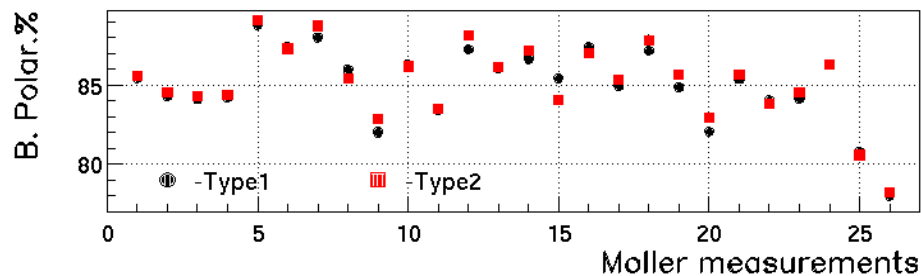


Type 2 measurement along foil 4

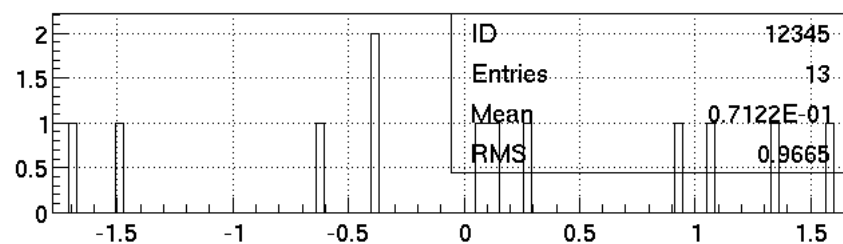
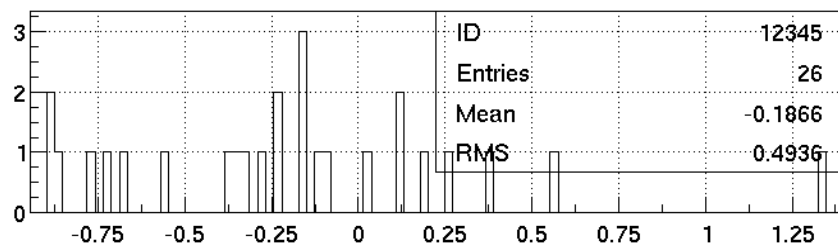
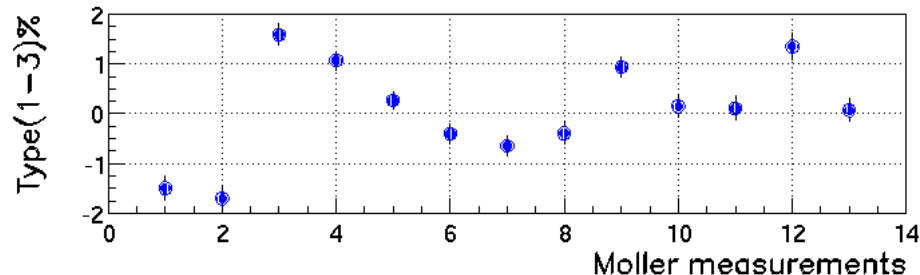
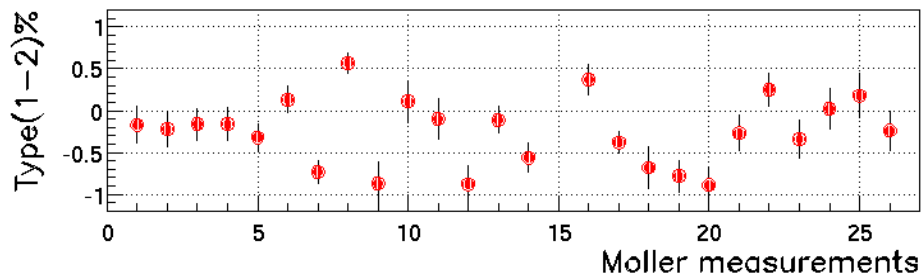
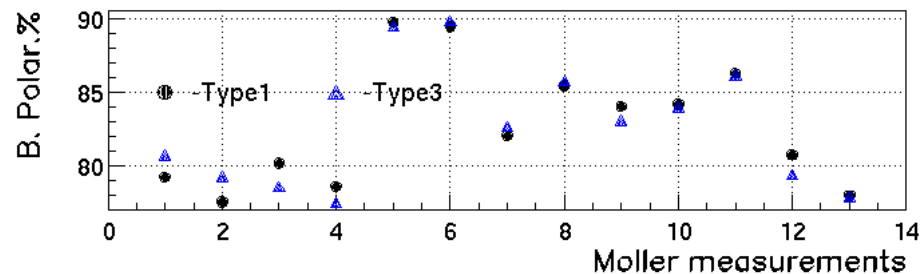


Difference between types

Difference between Type 1 and 2



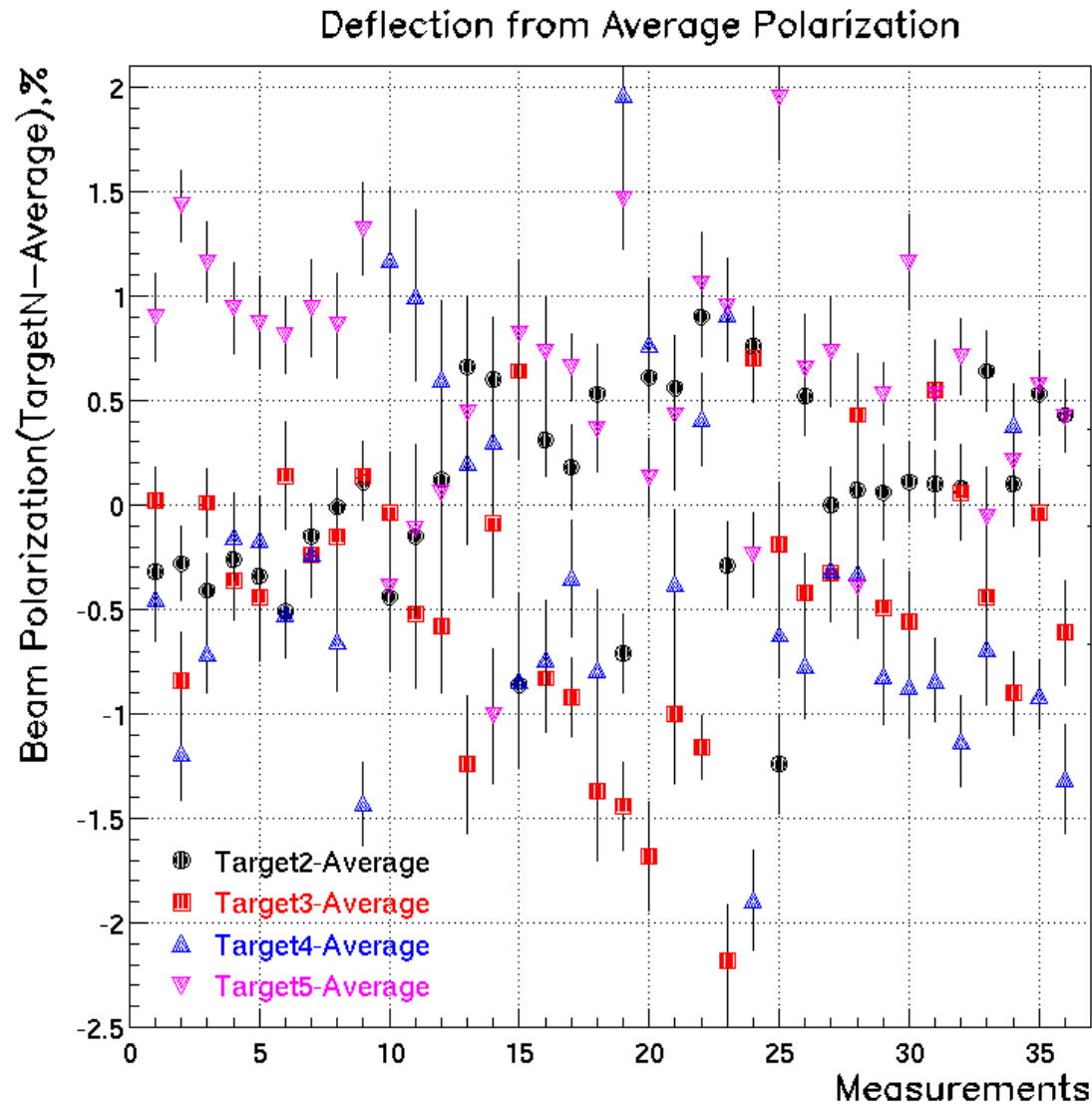
Difference between Type 1 and 3



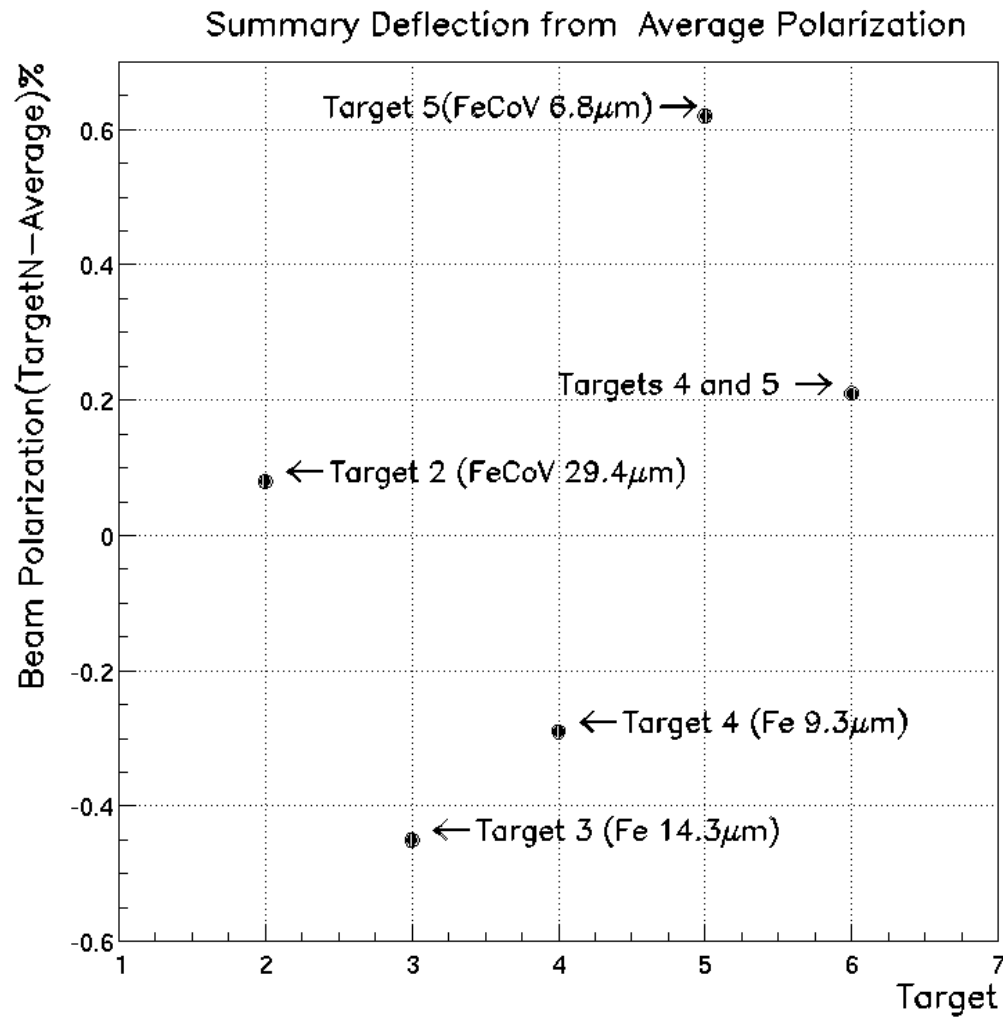
*Systematic error Type(1-2):
(0.2+0.5)=±0.7%*

*Systematic error Type(1-3):
±1.0%*

Beam polarization measurement with all four targets



Systematics for different targets



Systematic error (all targets): $(0.6+0.4)/2=\pm 0.5\%$

Target Systematic Error

<i>Variable</i>	<i>Error, %</i>
<i>Stand measurements</i>	<i>0.23</i>
<i>Double Scan (Left vs. Right)</i>	<i>0.3</i>
<i>Type 1 vs. Type 2</i>	<i>0.7</i>
<i>Type 1 vs. Type 3</i>	<i>1.0</i>
<i>All Targets</i>	<i>0.5</i>
<i>Target angle</i>	<i>0.3</i>
<i>Others</i>	<i>0.5</i>
<i>Total</i>	<i>1.5</i>

Møller Polarimeter Systematic Error (2005-2009)

<i>Variable</i>	<i>Error, %</i>
<i>Target</i>	<i>1.5</i>
<i>Analyzing Power</i>	<i>0.3</i>
<i>Levchuk-effect</i>	<i>0.2</i>
<i>Dead time</i>	<i>0.3</i>
<i>Background</i>	<i>0.3</i>
<i>Others</i>	<i>0.5</i>
<i>Total</i>	<i>1.7</i>

Møller Polarimeter Systematic Error (2005-2009)*

<i>Variable</i>	<i>Error, %</i>
<i>Target</i>	<i>1.5</i>
<i>Analyzing power</i>	<i>0.3</i>
<i>Levchuk-effect</i>	<i>0.2</i>
<i>Dead time</i>	<i>0.3</i>
<i>Background</i>	<i>0.3</i>
<i>Beam energy fluctuation</i>	<i>1.0</i>
<i>Others</i>	<i>0.5</i>
<i>Total</i>	<i>2.0</i>

*- for experiments with polarization correction
on the beam energy fluctuation